

PU5558-PU5567-computer-lab-report

Anonymous

Load necessary packages

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2     3.5.1      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(tidymodels)
```

```
## -- Attaching packages ----- tidymodels 1.2.0 --
## v broom        1.0.7      v rsample     1.2.1
## v dials         1.4.0      v tune        1.2.1
## v infer         1.0.7      v workflows   1.2.0
## v modeldata     1.4.0      v workflowsets 1.1.0
## v parsnip       1.3.0      v yardstick   1.3.2
## v recipes       1.1.1
## -- Conflicts ----- tidymodels_conflicts() --
## x scales::discard() masks purrr::discard()
## x dplyr::filter()   masks stats::filter()
## x recipes::fixed() masks stringr::fixed()
## x dplyr::lag()      masks stats::lag()
## x yardstick::spec() masks readr::spec()
## x recipes::step()   masks stats::step()
## * Search for functions across packages at https://www.tidymodels.org/find/
```

```
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
library(psych)
```

```
## Warning: package 'psych' was built under R version 4.4.3
##
## Attaching package: 'psych'
##
## The following objects are masked from 'package:scales':
##
##   alpha, rescale
```

```
##
## The following objects are masked from 'package:ggplot2':
##
##      %+%, alpha
```

```
library(vip)
```

```
##
## Attaching package: 'vip'
##
## The following object is masked from 'package:utils':
##
##      vi
```

Load chosen dataset

```
kneeData <- read_csv("Knee_Replacement_CCG 2021.csv")
```

```
## Rows: 5422 Columns: 81
## -- Column specification -----
## Delimiter: ","
## chr (5): Provider Code, Procedure, Year, Age Band, Gender
## dbl (76): Revision Flag, Pre-Op Q Assisted, Pre-Op Q Assisted By, Pre-Op Q S...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
kneeData <- kneeData %>%
  select(-Procedure, -Year, -"Revision Flag", -Gender, -"Age Band", -"Provider Code") %>%
  drop_na() %>%
  unique()
glimpse(kneeData)
```

```
## Rows: 4,528
## Columns: 75
## $ `Pre-Op Q Assisted`      <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Pre-Op Q Assisted By`  <dbl> 0, 0, 0, 0, 0, 0, 0~
## $ `Pre-Op Q Symptom Period` <dbl> 2, 4, 4, 2, 4, 2, 2~
## $ `Pre-Op Q Previous Surgery` <dbl> 2, 1, 2, 2, 2, 2, 2~
## $ `Pre-Op Q Living Arrangements` <dbl> 2, 2, 1, 1, 1, 1, 1~
## $ `Pre-Op Q Disability`    <dbl> 1, 2, 2, 2, 1, 2, 2~
## $ `Heart Disease`         <dbl> 9, 9, 9, 9, 9, 1, 9~
## $ `High Bp`               <dbl> 1, 9, 9, 1, 1, 1, 9~
## $ Stroke                  <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ Circulation              <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ `Lung Disease`          <dbl> 9, 9, 9, 1, 9, 9, 9~
## $ Diabetes                 <dbl> 9, 9, 9, 1, 9, 9, 9~
## $ `Kidney Disease`        <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ `Nervous System`        <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ `Liver Disease`         <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ Cancer                  <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ Depression              <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ Arthritis               <dbl> 9, 1, 1, 1, 1, 1, 1~
## $ `Pre-Op Q Mobility`     <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Pre-Op Q Self-Care`    <dbl> 1, 1, 1, 1, 1, 1, 1~
```

```

## $ `Pre-Op Q Activity` <dbl> 3, 2, 3, 2, 1, 2, 2~
## $ `Pre-Op Q Discomfort` <dbl> 2, 2, 3, 2, 1, 2, 2~
## $ `Pre-Op Q Anxiety` <dbl> 1, 1, 2, 2, 1, 2, 1~
## $ `Pre-Op Q EQ5D Index Profile` <dbl> 21321, 21221, 21332~
## $ `Pre-Op Q EQ5D Index` <dbl> 0.364, 0.691, 0.030~
## $ `Post-Op Q Assisted` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q Assisted By` <dbl> 9, 9, 9, 9, 9, 9, 9~
## $ `Post-Op Q Living Arrangements` <dbl> 2, 2, 1, 1, 1, 1, 1~
## $ `Post-Op Q Disability` <dbl> 1, 2, 2, 2, 2, 1, 2~
## $ `Post-Op Q Mobility` <dbl> 1, 1, 1, 1, 1, 2, 1~
## $ `Post-Op Q Self-Care` <dbl> 2, 1, 1, 1, 1, 2, 1~
## $ `Post-Op Q Activity` <dbl> 1, 1, 1, 1, 1, 2, 1~
## $ `Post-Op Q Discomfort` <dbl> 1, 1, 1, 1, 1, 2, 2~
## $ `Post-Op Q Anxiety` <dbl> 2, 1, 1, 1, 1, 1, 1~
## $ `Post-Op Q Satisfaction` <dbl> 2, 3, 1, 2, 1, 4, 3~
## $ `Post-Op Q Success` <dbl> 1, 1, 1, 1, 1, 4, 1~
## $ `Post-Op Q Allergy` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q Bleeding` <dbl> 2, 2, 2, 2, 1, 2, 2~
## $ `Post-Op Q Wound` <dbl> 1, 2, 2, 2, 1, 2, 2~
## $ `Post-Op Q Urine` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q Further Surgery` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q Readmitted` <dbl> 2, 2, 2, 2, 2, 2, 2~
## $ `Post-Op Q EQ5D Index Profile` <dbl> 12112, 11111, 11111~
## $ `Post-Op Q EQ5D Index` <dbl> 0.744, 1.000, 1.000~
## $ `Knee Replacement EQ 5D Index Post-Op Q Predicted` <dbl> 0.6621424, 0.740953~
## $ `Pre-Op Q EQ VAS` <dbl> 85, 50, 46, 60, 60,~
## $ `Post-Op Q EQ VAS` <dbl> 60, 100, 90, 95, 90~
## $ `Knee Replacement EQ VAS_Post-Op Q Predicted` <dbl> 75.63073, 66.38606,~
## $ `Knee Replacement Pre-Op Q Pain` <dbl> 1, 1, 0, 0, 1, 1, 1~
## $ `Knee Replacement Pre-Op Q Night Pain` <dbl> 1, 2, 2, 2, 2, 3, 0~
## $ `Knee Replacement Pre-Op Q Washing` <dbl> 3, 4, 3, 4, 3, 3, 3~
## $ `Knee Replacement Pre-Op Q Transport` <dbl> 1, 4, 2, 4, 2, 2, 2~
## $ `Knee Replacement Pre-Op Q Walking` <dbl> 3, 3, 2, 3, 3, 1, 2~
## $ `Knee Replacement Pre-Op Q Standing` <dbl> 3, 2, 2, 3, 3, 2, 2~
## $ `Knee Replacement Pre-Op Q Limping` <dbl> 0, 3, 0, 0, 3, 1, 1~
## $ `Knee Replacement Pre-Op Q Kneeling` <dbl> 0, 0, 0, 2, 1, 1, 2~
## $ `Knee Replacement Pre-Op Q Work` <dbl> 2, 2, 1, 3, 2, 1, 2~
## $ `Knee Replacement Pre-Op Q Confidence` <dbl> 3, 3, 3, 2, 1, 1, 2~
## $ `Knee Replacement Pre-Op Q Shopping` <dbl> 2, 4, 0, 2, 3, 0, 2~
## $ `Knee Replacement Pre-Op Q Stairs` <dbl> 1, 4, 2, 2, 3, 2, 2~
## $ `Knee Replacement Pre-Op Q Score` <dbl> 20, 32, 17, 27, 27,~
## $ `Knee Replacement Post-Op Q Pain` <dbl> 4, 3, 3, 4, 3, 1, 2~
## $ `Knee Replacement Post-Op Q Night Pain` <dbl> 4, 4, 3, 4, 4, 3, 2~
## $ `Knee Replacement Post-Op Q Washing` <dbl> 3, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Transport` <dbl> 2, 4, 4, 4, 4, 3, 4~
## $ `Knee Replacement Post-Op Q Walking` <dbl> 4, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Standing` <dbl> 3, 4, 4, 4, 4, 1, 3~
## $ `Knee Replacement Post-Op Q Limping` <dbl> 3, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Kneeling` <dbl> 0, 4, 0, 3, 2, 0, 2~
## $ `Knee Replacement Post-Op Q Work` <dbl> 3, 3, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Confidence` <dbl> 4, 4, 4, 4, 4, 1, 4~
## $ `Knee Replacement Post-Op Q Shopping` <dbl> 4, 4, 4, 4, 4, 0, 4~
## $ `Knee Replacement Post-Op Q Stairs` <dbl> 3, 4, 4, 4, 3, 1, 4~
## $ `Knee Replacement Post-Op Q Score` <dbl> 37, 46, 42, 47, 44,~

```

```
## $ `Knee Replacement OKS Post-Op Q Predicted` <dbl> 34.02619, 36.88715,~
```

Dataset description

```
describe(kneeData, fast=TRUE)
```

##	vars	n	mean	sd
## Pre-Op Q Assisted	1	4528	2.00	0.98
## Pre-Op Q Assisted By	2	4528	0.00	0.00
## Pre-Op Q Symptom Period	3	4528	2.70	1.01
## Pre-Op Q Previous Surgery	4	4528	1.95	0.66
## Pre-Op Q Living Arrangements	5	4528	1.29	0.90
## Pre-Op Q Disability	6	4528	1.85	1.55
## Heart Disease	7	4528	8.35	2.19
## High Bp	8	4528	5.62	3.95
## Stroke	9	4528	8.91	0.86
## Circulation	10	4528	8.68	1.56
## Lung Disease	11	4528	8.32	2.23
## Diabetes	12	4528	8.19	2.41
## Kidney Disease	13	4528	8.85	1.09
## Nervous System	14	4528	8.90	0.88
## Liver Disease	15	4528	8.93	0.73
## Cancer	16	4528	8.57	1.81
## Depression	17	4528	8.28	2.29
## Arthritis	18	4528	2.95	3.44
## Pre-Op Q Mobility	19	4528	1.90	0.31
## Pre-Op Q Self-Care	20	4528	1.33	0.48
## Pre-Op Q Activity	21	4528	2.02	0.49
## Pre-Op Q Discomfort	22	4528	2.32	0.50
## Pre-Op Q Anxiety	23	4528	1.41	0.56
## Pre-Op Q EQ5D Index Profile	24	4528	20543.07	3241.11
## Pre-Op Q EQ5D Index	25	4528	0.44	0.31
## Post-Op Q Assisted	26	4528	1.98	0.56
## Post-Op Q Assisted By	27	4528	8.50	1.91
## Post-Op Q Living Arrangements	28	4528	1.29	0.84
## Post-Op Q Disability	29	4528	1.83	1.13
## Post-Op Q Mobility	30	4528	1.43	0.50
## Post-Op Q Self-Care	31	4528	1.16	0.38
## Post-Op Q Activity	32	4528	1.53	0.56
## Post-Op Q Discomfort	33	4528	1.65	0.56
## Post-Op Q Anxiety	34	4528	1.24	0.48
## Post-Op Q Satisfaction	35	4528	2.26	1.14
## Post-Op Q Success	36	4528	1.43	0.92
## Post-Op Q Allergy	37	4528	2.44	1.89
## Post-Op Q Bleeding	38	4528	2.64	2.13
## Post-Op Q Wound	39	4528	2.48	1.99
## Post-Op Q Urine	40	4528	2.52	2.04
## Post-Op Q Further Surgery	41	4528	1.99	0.33
## Post-Op Q Readmitted	42	4528	1.95	0.40
## Post-Op Q EQ5D Index Profile	43	4528	15623.63	5175.67
## Post-Op Q EQ5D Index	44	4528	0.75	0.24
## Knee Replacement EQ 5D Index Post-Op Q Predicted	45	4528	0.74	0.12
## Pre-Op Q EQ VAS	46	4528	67.87	20.22
## Post-Op Q EQ VAS	47	4528	75.37	17.58

## Knee Replacement EQ VAS_Post-Op Q Predicted	48 4528	73.27	9.30
## Knee Replacement Pre-Op Q Pain	49 4528	0.63	0.67
## Knee Replacement Pre-Op Q Night Pain	50 4528	1.27	1.15
## Knee Replacement Pre-Op Q Washing	51 4528	2.81	0.98
## Knee Replacement Pre-Op Q Transport	52 4528	2.18	0.88
## Knee Replacement Pre-Op Q Walking	53 4528	2.09	1.08
## Knee Replacement Pre-Op Q Standing	54 4528	1.73	0.87
## Knee Replacement Pre-Op Q Limping	55 4528	0.95	1.05
## Knee Replacement Pre-Op Q Kneeling	56 4528	0.88	0.89
## Knee Replacement Pre-Op Q Work	57 4528	1.48	0.90
## Knee Replacement Pre-Op Q Confidence	58 4528	1.95	1.20
## Knee Replacement Pre-Op Q Shopping	59 4528	1.88	1.21
## Knee Replacement Pre-Op Q Stairs	60 4528	1.82	0.93
## Knee Replacement Pre-Op Q Score	61 4528	19.68	7.99
## Knee Replacement Post-Op Q Pain	62 4528	2.62	1.11
## Knee Replacement Post-Op Q Night Pain	63 4528	2.83	1.25
## Knee Replacement Post-Op Q Washing	64 4528	3.61	0.70
## Knee Replacement Post-Op Q Transport	65 4528	3.15	0.87
## Knee Replacement Post-Op Q Walking	66 4528	3.40	0.94
## Knee Replacement Post-Op Q Standing	67 4528	3.15	0.86
## Knee Replacement Post-Op Q Limping	68 4528	3.12	1.07
## Knee Replacement Post-Op Q Kneeling	69 4528	1.66	1.32
## Knee Replacement Post-Op Q Work	70 4528	3.08	0.98
## Knee Replacement Post-Op Q Confidence	71 4528	3.53	0.80
## Knee Replacement Post-Op Q Shopping	72 4528	3.19	1.18
## Knee Replacement Post-Op Q Stairs	73 4528	3.06	1.01
## Knee Replacement Post-Op Q Score	74 4528	36.38	9.11
## Knee Replacement OKS Post-Op Q Predicted	75 4528	34.90	4.74
##	median	min	max
## Pre-Op Q Assisted	2.00	1.00	9.00
## Pre-Op Q Assisted By	0.00	0.00	0.00
## Pre-Op Q Symptom Period	2.00	1.00	9.00
## Pre-Op Q Previous Surgery	2.00	1.00	9.00
## Pre-Op Q Living Arrangements	1.00	1.00	9.00
## Pre-Op Q Disability	2.00	1.00	9.00
## Heart Disease	9.00	1.00	9.00
## High Bp	9.00	1.00	9.00
## Stroke	9.00	1.00	9.00
## Circulation	9.00	1.00	9.00
## Lung Disease	9.00	1.00	9.00
## Diabetes	9.00	1.00	9.00
## Kidney Disease	9.00	1.00	9.00
## Nervous System	9.00	1.00	9.00
## Liver Disease	9.00	1.00	9.00
## Cancer	9.00	1.00	9.00
## Depression	9.00	1.00	9.00
## Arthritis	1.00	1.00	9.00
## Pre-Op Q Mobility	2.00	1.00	3.00
## Pre-Op Q Self-Care	1.00	1.00	3.00
## Pre-Op Q Activity	2.00	1.00	3.00
## Pre-Op Q Discomfort	2.00	1.00	3.00
## Pre-Op Q Anxiety	1.00	1.00	3.00
## Pre-Op Q EQ5D Index Profile	21222.00	11111.00	33332.00
## Pre-Op Q EQ5D Index	0.59	-0.43	1.00

## Post-Op Q Assisted	2.00	1.00	9.00	
## Post-Op Q Assisted By	9.00	1.00	9.00	
## Post-Op Q Living Arrangements	1.00	1.00	9.00	
## Post-Op Q Disability	2.00	1.00	9.00	
## Post-Op Q Mobility	1.00	1.00	3.00	
## Post-Op Q Self-Care	1.00	1.00	3.00	
## Post-Op Q Activity	1.00	1.00	3.00	
## Post-Op Q Discomfort	2.00	1.00	3.00	
## Post-Op Q Anxiety	1.00	1.00	3.00	
## Post-Op Q Satisfaction	2.00	1.00	9.00	
## Post-Op Q Success	1.00	1.00	9.00	
## Post-Op Q Allergy	2.00	1.00	9.00	
## Post-Op Q Bleeding	2.00	1.00	9.00	
## Post-Op Q Wound	2.00	1.00	9.00	
## Post-Op Q Urine	2.00	1.00	9.00	
## Post-Op Q Further Surgery	2.00	1.00	9.00	
## Post-Op Q Readmitted	2.00	1.00	9.00	
## Post-Op Q EQ5D Index Profile	11221.00	11111.00	33333.00	
## Post-Op Q EQ5D Index	0.76	-0.59	1.00	
## Knee Replacement EQ 5D Index Post-Op Q Predicted	0.76	0.03	0.97	
## Pre-Op Q EQ VAS	70.00	0.00	100.00	
## Post-Op Q EQ VAS	80.00	0.00	100.00	
## Knee Replacement EQ VAS_Post-Op Q Predicted	73.85	37.62	93.48	
## Knee Replacement Pre-Op Q Pain	1.00	0.00	4.00	
## Knee Replacement Pre-Op Q Night Pain	1.00	0.00	4.00	
## Knee Replacement Pre-Op Q Washing	3.00	0.00	4.00	
## Knee Replacement Pre-Op Q Transport	2.00	0.00	4.00	
## Knee Replacement Pre-Op Q Walking	2.00	0.00	4.00	
## Knee Replacement Pre-Op Q Standing	2.00	0.00	4.00	
## Knee Replacement Pre-Op Q Limping	1.00	0.00	4.00	
## Knee Replacement Pre-Op Q Kneeling	1.00	0.00	4.00	
## Knee Replacement Pre-Op Q Work	1.00	0.00	4.00	
## Knee Replacement Pre-Op Q Confidence	2.00	0.00	4.00	
## Knee Replacement Pre-Op Q Shopping	2.00	0.00	4.00	
## Knee Replacement Pre-Op Q Stairs	2.00	0.00	4.00	
## Knee Replacement Pre-Op Q Score	19.00	0.00	46.00	
## Knee Replacement Post-Op Q Pain	3.00	0.00	4.00	
## Knee Replacement Post-Op Q Night Pain	3.00	0.00	4.00	
## Knee Replacement Post-Op Q Washing	4.00	0.00	4.00	
## Knee Replacement Post-Op Q Transport	3.00	0.00	4.00	
## Knee Replacement Post-Op Q Walking	4.00	0.00	4.00	
## Knee Replacement Post-Op Q Standing	3.00	0.00	4.00	
## Knee Replacement Post-Op Q Limping	3.00	0.00	4.00	
## Knee Replacement Post-Op Q Kneeling	2.00	0.00	4.00	
## Knee Replacement Post-Op Q Work	3.00	0.00	4.00	
## Knee Replacement Post-Op Q Confidence	4.00	0.00	4.00	
## Knee Replacement Post-Op Q Shopping	4.00	0.00	4.00	
## Knee Replacement Post-Op Q Stairs	3.00	0.00	4.00	
## Knee Replacement Post-Op Q Score	38.00	3.00	48.00	
## Knee Replacement OKS Post-Op Q Predicted	35.54	7.95	44.79	
##	range	skew	kurtosis	se
## Pre-Op Q Assisted	8.00	6.15	42.16	0.01
## Pre-Op Q Assisted By	0.00	NaN	NaN	0.00
## Pre-Op Q Symptom Period	8.00	2.09	9.61	0.02

## Pre-Op Q Previous Surgery	8.00	8.22	88.95	0.01
## Pre-Op Q Living Arrangements	8.00	6.71	53.29	0.01
## Pre-Op Q Disability	8.00	3.91	15.33	0.02
## Heart Disease	8.00	-3.06	7.36	0.03
## High Bp	8.00	-0.32	-1.90	0.06
## Stroke	8.00	-9.08	80.41	0.01
## Circulation	8.00	-4.71	20.19	0.02
## Lung Disease	8.00	-2.97	6.85	0.03
## Diabetes	8.00	-2.64	5.00	0.04
## Kidney Disease	8.00	-7.05	47.65	0.02
## Nervous System	8.00	-8.90	77.30	0.01
## Liver Disease	8.00	-10.77	114.11	0.01
## Cancer	8.00	-3.94	13.53	0.03
## Depression	8.00	-2.86	6.19	0.03
## Arthritis	8.00	1.19	-0.58	0.05
## Pre-Op Q Mobility	2.00	-2.42	4.79	0.00
## Pre-Op Q Self-Care	2.00	0.91	-0.71	0.01
## Pre-Op Q Activity	2.00	0.06	1.22	0.01
## Pre-Op Q Discomfort	2.00	0.37	-0.99	0.01
## Pre-Op Q Anxiety	2.00	0.98	-0.06	0.01
## Pre-Op Q EQ5D Index Profile	22221.00	-2.30	4.50	48.17
## Pre-Op Q EQ5D Index	1.43	-0.52	-1.27	0.00
## Post-Op Q Assisted	8.00	9.92	124.89	0.01
## Post-Op Q Assisted By	8.00	-3.60	11.07	0.03
## Post-Op Q Living Arrangements	8.00	6.93	59.70	0.01
## Post-Op Q Disability	8.00	5.06	29.94	0.02
## Post-Op Q Mobility	2.00	0.31	-1.83	0.01
## Post-Op Q Self-Care	2.00	2.19	3.73	0.01
## Post-Op Q Activity	2.00	0.46	-0.81	0.01
## Post-Op Q Discomfort	2.00	0.12	-0.76	0.01
## Post-Op Q Anxiety	2.00	1.76	2.24	0.01
## Post-Op Q Satisfaction	8.00	1.29	4.07	0.02
## Post-Op Q Success	8.00	3.01	12.41	0.01
## Post-Op Q Allergy	8.00	3.10	7.96	0.03
## Post-Op Q Bleeding	8.00	2.61	4.96	0.03
## Post-Op Q Wound	8.00	2.87	6.60	0.03
## Post-Op Q Urine	8.00	2.79	6.07	0.03
## Post-Op Q Further Surgery	8.00	15.74	343.24	0.00
## Post-Op Q Readmitted	8.00	9.89	187.99	0.01
## Post-Op Q EQ5D Index Profile	22222.00	0.33	-1.78	76.92
## Post-Op Q EQ5D Index	1.59	-1.50	2.87	0.00
## Knee Replacement EQ 5D Index Post-Op Q Predicted	0.94	-0.90	0.88	0.00
## Pre-Op Q EQ VAS	100.00	-0.67	-0.09	0.30
## Post-Op Q EQ VAS	100.00	-1.15	1.39	0.26
## Knee Replacement EQ VAS_Post-Op Q Predicted	55.86	-0.37	-0.18	0.14
## Knee Replacement Pre-Op Q Pain	4.00	1.01	1.82	0.01
## Knee Replacement Pre-Op Q Night Pain	4.00	0.72	-0.14	0.02
## Knee Replacement Pre-Op Q Washing	4.00	-0.26	-0.90	0.01
## Knee Replacement Pre-Op Q Transport	4.00	0.43	-0.21	0.01
## Knee Replacement Pre-Op Q Walking	4.00	-0.26	-0.46	0.02
## Knee Replacement Pre-Op Q Standing	4.00	0.41	-0.25	0.01
## Knee Replacement Pre-Op Q Limping	4.00	0.95	0.00	0.02
## Knee Replacement Pre-Op Q Kneeling	4.00	0.77	0.04	0.01
## Knee Replacement Pre-Op Q Work	4.00	0.34	-0.04	0.01

## Knee Replacement Pre-Op Q Confidence	4.00	-0.01	-0.99	0.02
## Knee Replacement Pre-Op Q Shopping	4.00	0.01	-0.86	0.02
## Knee Replacement Pre-Op Q Stairs	4.00	0.29	-0.16	0.01
## Knee Replacement Pre-Op Q Score	46.00	0.25	-0.35	0.12
## Knee Replacement Post-Op Q Pain	4.00	-0.40	-0.81	0.02
## Knee Replacement Post-Op Q Night Pain	4.00	-0.73	-0.56	0.02
## Knee Replacement Post-Op Q Washing	4.00	-1.87	3.07	0.01
## Knee Replacement Post-Op Q Transport	4.00	-0.66	-0.41	0.01
## Knee Replacement Post-Op Q Walking	4.00	-1.71	2.52	0.01
## Knee Replacement Post-Op Q Standing	4.00	-0.78	-0.03	0.01
## Knee Replacement Post-Op Q Limping	4.00	-1.36	1.23	0.02
## Knee Replacement Post-Op Q Kneeling	4.00	0.14	-1.23	0.02
## Knee Replacement Post-Op Q Work	4.00	-0.91	0.22	0.01
## Knee Replacement Post-Op Q Confidence	4.00	-2.05	4.44	0.01
## Knee Replacement Post-Op Q Shopping	4.00	-1.45	1.14	0.02
## Knee Replacement Post-Op Q Stairs	4.00	-0.93	0.25	0.02
## Knee Replacement Post-Op Q Score	45.00	-0.97	0.44	0.14
## Knee Replacement OKS Post-Op Q Predicted	36.84	-0.85	0.92	0.07

The following dataset depicts ## Suitable machine learning algorithm for three questions:

1. Before the operation, can we estimate the post-operative EQ5D index for a patient? Given certain presence of certain predictors being present, this would fall into supervised learning there are no missing values so no semi supervised is needed 0.364, 0.691, can use simple linear regression then test how well the model preforms with different AIC values. This question will be used for this report
2. Before the operation, can we predict how much pain a patient will have after the operation? Since pain is measured in categories we would want a classifier to predict which index the post operative pain will be in this case we would use an unsupervised algorithm, seems intuitive that you would use forests

0 = Severe 1 = Moderate 2 = Mild 3 = Very Mild 4 = None 9 = Missing

3. Before the operation, can we calculate how many patients have had previous surgery? Here we can use supervised/semi-supervised learning methods to cualcyate the numerical value of the patiens that have had the previous surgery. The results is binary wether they have or have not had it

Model building to answer chosen question

1. Data splitting

```
kneeSplit <- kneeData %>%
  initial_split(prop = 0.8, strata = "Post-Op Q EQ5D Index")
kneeTrain <- training(kneeSplit)
kneeTest <- testing(kneeSplit)
print(sapply(kneeTrain, is.numeric))
```

```
##               Pre-Op Q Assisted
##                      TRUE
##           Pre-Op Q Assisted By
##                      TRUE
##           Pre-Op Q Symptom Period
##                      TRUE
##           Pre-Op Q Previous Surgery
##                      TRUE
##           Pre-Op Q Living Arrangements
##                      TRUE
##           Pre-Op Q Disability
```


##	TRUE
##	Heart Disease
##	TRUE
##	High Bp
##	TRUE
##	Stroke
##	TRUE
##	Circulation
##	TRUE
##	Lung Disease
##	TRUE
##	Diabetes
##	TRUE
##	Kidney Disease
##	TRUE
##	Nervous System
##	TRUE
##	Liver Disease
##	TRUE
##	Cancer
##	TRUE
##	Depression
##	TRUE
##	Arthritis
##	TRUE
##	Pre-Op Q Mobility
##	TRUE
##	Pre-Op Q Self-Care
##	TRUE
##	Pre-Op Q Activity
##	TRUE
##	Pre-Op Q Discomfort
##	TRUE
##	Pre-Op Q Anxiety
##	TRUE
##	Pre-Op Q EQ5D Index Profile
##	TRUE
##	Pre-Op Q EQ5D Index
##	TRUE
##	Post-Op Q Assisted
##	TRUE
##	Post-Op Q Assisted By
##	TRUE
##	Post-Op Q Living Arrangements
##	TRUE
##	Post-Op Q Disability
##	TRUE
##	Post-Op Q Mobility
##	TRUE
##	Post-Op Q Self-Care
##	TRUE
##	Post-Op Q Activity
##	TRUE
##	Post-Op Q Discomfort

```

##                                     TRUE
##                               Post-Op Q Anxiety
##                                     TRUE
##                               Post-Op Q Satisfaction
##                                     TRUE
##                               Post-Op Q Success
##                                     TRUE
##                               Post-Op Q Allergy
##                                     TRUE
##                               Post-Op Q Bleeding
##                                     TRUE
##                               Post-Op Q Wound
##                                     TRUE
##                               Post-Op Q Urine
##                                     TRUE
##                               Post-Op Q Further Surgery
##                                     TRUE
##                               Post-Op Q Readmitted
##                                     TRUE
##                               Post-Op Q EQ5D Index Profile
##                                     TRUE
##                               Post-Op Q EQ5D Index
##                                     TRUE
## Knee Replacement EQ 5D Index Post-Op Q Predicted
##                                     TRUE
##                               Pre-Op Q EQ VAS
##                                     TRUE
##                               Post-Op Q EQ VAS
##                                     TRUE
## Knee Replacement EQ VAS_Post-Op Q Predicted
##                                     TRUE
##                               Knee Replacement Pre-Op Q Pain
##                                     TRUE
##                               Knee Replacement Pre-Op Q Night Pain
##                                     TRUE
##                               Knee Replacement Pre-Op Q Washing
##                                     TRUE
##                               Knee Replacement Pre-Op Q Transport
##                                     TRUE
##                               Knee Replacement Pre-Op Q Walking
##                                     TRUE
##                               Knee Replacement Pre-Op Q Standing
##                                     TRUE
##                               Knee Replacement Pre-Op Q Limping
##                                     TRUE
##                               Knee Replacement Pre-Op Q Kneeling
##                                     TRUE
##                               Knee Replacement Pre-Op Q Work
##                                     TRUE
##                               Knee Replacement Pre-Op Q Confidence
##                                     TRUE
##                               Knee Replacement Pre-Op Q Shopping
##                                     TRUE
##                               Knee Replacement Pre-Op Q Stairs

```

```
##                                TRUE
##      Knee Replacement Pre-Op Q Score
##                                TRUE
##      Knee Replacement Post-Op Q Pain
##                                TRUE
##      Knee Replacement Post-Op Q Night Pain
##                                TRUE
##      Knee Replacement Post-Op Q Washing
##                                TRUE
##      Knee Replacement Post-Op Q Transport
##                                TRUE
##      Knee Replacement Post-Op Q Walking
##                                TRUE
##      Knee Replacement Post-Op Q Standing
##                                TRUE
##      Knee Replacement Post-Op Q Limping
##                                TRUE
##      Knee Replacement Post-Op Q Kneeling
##                                TRUE
##      Knee Replacement Post-Op Q Work
##                                TRUE
##      Knee Replacement Post-Op Q Confidence
##                                TRUE
##      Knee Replacement Post-Op Q Shopping
##                                TRUE
##      Knee Replacement Post-Op Q Stairs
##                                TRUE
##      Knee Replacement Post-Op Q Score
##                                TRUE
##      Knee Replacement OKS Post-Op Q Predicted
##                                TRUE
```

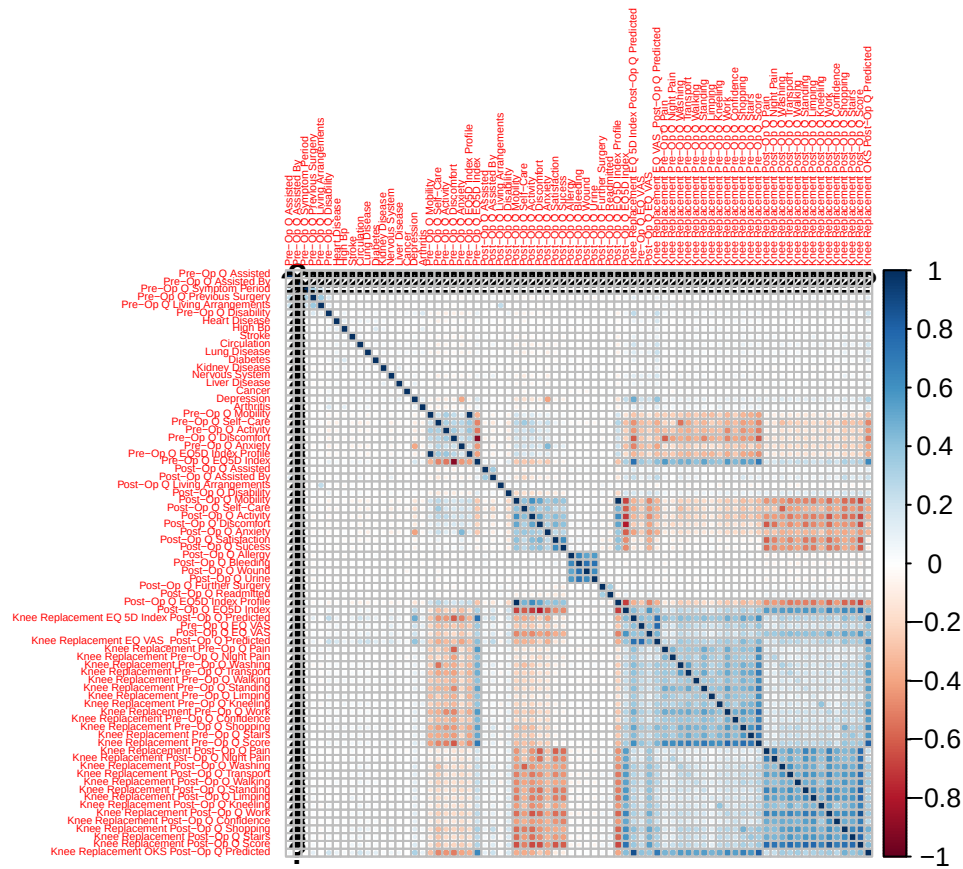
2. Selection and preprocessing of predictors There are different methods that can be used to address using predictors that are significant to the model the following will be explored

1: Correlation plots 2: Stepwise Regression

```
kneeCorr <- cor(kneeTrain)
```

```
## Warning in cor(kneeTrain): the standard deviation is zero
```

```
corrplot(kneeCorr, tl.cex=0.3)
```



3. Model specification and training

4. Model evaluation

Limitations of machine learning model